Innovation Report

Report overview

This project includes an additional reporting and analytics feature implemented as Option 7 in the conference management application.

The feature generates an automated attendance analysis report using live conference registration data stored in the MySQL database.

Instead of only displaying records in the terminal, the system creates:

- a PDF attendance report
- a generated attendance graph
- automatic attendance analysis

This adds more practical use to the application by turning database records into reusable reports and analytics.

Purpose of the Feature

The purpose of the attendance reporting feature is to help conference organisers:

- monitor session attendance
- review room usage
- check available seating capacity
- identify highly attended sessions
- support conference planning decisions

The reporting system automatically updates whenever database records change.

If new registrations are added to the database, the generated report and graph are recreated using the newest/most recent available data.

Technologies and Libraries Used

The feature was implemented using:

Technology	Purpose
MySQL	Conference data storage
Python	Backend application logic
Matplotlib	Attendance graph generation
ReportLab	PDF report generation
SQL Aggregation	Attendance calculations and analytics

SQL and Analytics Logic

The attendance report uses SQL aggregation functions and JOIN operations to combine:

- room data
- session data
- registration data

The report calculates:

- registered attendees per session
- available room seats
- attendance percentage
- average attendance
- highest and lowest attendance sessions

The attendance percentage is calculated using:

(registered attendees / room capacity) * 100

This allows the report to measure room occupancy rather than only raw attendee numbers.

Generated Report and Graph

The application automatically generates:

- generated_attendance_report.pdf
- generated_attendance_graph.png

The PDF report contains:

- attendance table
- attendance percentage graph
- automatic analysis section

The graph visualises room attendance percentages using matplotlib.

Innovation and Business Value

This feature adds innovation to the project because the application:

- automates report generation
- creates graphical attendance analysis
- exports reusable PDF reports
- dynamically updates reporting outputs from database changes

The reporting system demonstrates how databases can support real-world business analytics. This feature demonstrates how relational data can be transformed into actionable insights using SQL aggregation and automated reporting.

Limitations and Future Improvements

As this is a university project, the reporting feature is currently based on sample conference data. This means the results are useful for demonstrating the logic, but they do not represent a real live conference environment.

At the moment, the feature generates a current attendance report based on the data available in the database. It does not yet include long-term tracking or historical comparison.

Due to the project scope and time limits, the following features were not implemented yet, but they could be added in future versions:

- attendee trends over time
- room forecasting based on registration growth
- attendance history for previous sessions or events
- dashboard integration for live visual reporting
- CSV or Excel export functionality

These additions would make the reporting system more progressive and useful for larger conference planning, where organisers need to compare attendance patterns, predict demand, and export data in different formats.

Attendance Reporting Process Flow

OPTION 7: ATTENDANCE REPORT GENERATION LOGIC

STEP	PROCESS	DETAILS / CALCULATIONS
Step 1	User selects Option 7 from main menu	User chooses to generate the Attendance Report.
Step 2	Call function	Python calls generate_attendance_by_room_report()
Step 3	Connect to MySQL database	Establish connection using mysql.connector
Step 4	Execute SQL query with JOINs	Join room table, session table and registration table
Step 5	Apply SQL aggregation	COUNT(registrationID) to get registered attendees per session per room.
Step 6	Calculate values	Registered attendees = COUNT(registrationID) Available seats = Room Capacity - Registered attendees Attendance % = (Registered attendees / Room Capacity) × 100
Step 7	Fetch results into Python	Retrieve query results and store in a list of dictionaries or DataFrame.
Step 8	Display attendance table in terminal	Print formatted table showing: Room, Session, Speaker, Date, Capacity, Registered, Available Seats, Attendance %
Step 9	Generate attendance percentage graph	Using matplotlib: X-axis: Sessions | Y-axis: Attendance Percentage (%)
Step 10	Save graph	Graph saved as generated_attendance_graph.png
Step 11	Perform automatic analysis	- Highest registered attendance - Lowest registered attendance - Highest room occupancy percentage - Total registrations - Average attendance percentage
Step 12	Generate PDF report using ReportLab	Create a PDF file and set up document structure.
Step 13	Insert content into PDF	- Report title - Introduction - Attendance table - Attendance graph - Automatic analysis
Step 14	Save PDF report	PDF saved as generated_attendance_report.pdf
Step 15	End	Report generation completed successfully.

This diagram summarise how Option 7 works, from the user selecting the menu option to the generation of the attendance graph and PDF reports.

Resources and References

Documentation and Technical Resources

Python Official Documentation	https://docs.python.org/3/
MySQL Documentation	https://dev.mysql.com/doc/refman/8.0/en/
Matplotlib Documentation	https://matplotlib.org/stable/users/index.html
ReportLab Documentation	https://docs.reportlab.com/

Learning Resources

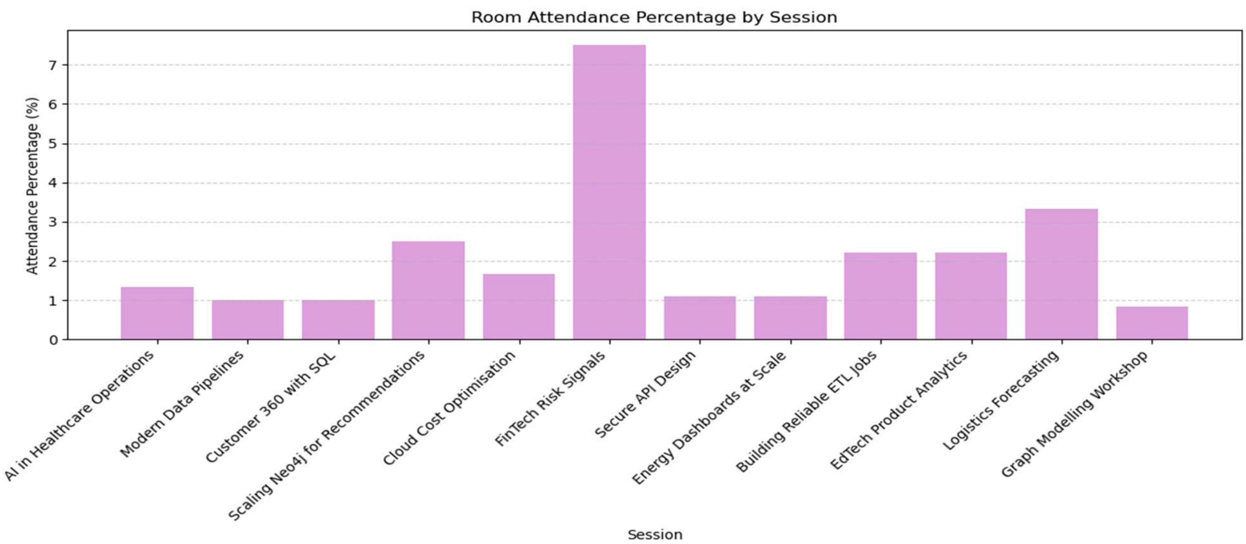
- Programming for Data Analytics module notes and class exercises
- Applied Databases module lecture materials at ATU
- SQL JOIN and aggregation examples demonstrated during ADB lectures labs

Libraries Used

- mysql-connector-python
- matplotlib
- reportlab

Appendices

Appendix A : Generated Attendance Percentage Graph



Appendix B : Generated Attendance Report Output

Conference Attendance Report

This report uses live registration data from the up-to-date database to show registered attendees, room capacity, available seats, and room attendance percentage.

Room	Session	Speaker	Date	Capacity	Registered	Available
Main Hall	AI in Healthcare Operations	Dr. Sara Khan	2025-05-13	300	4	296
Main Hall	Modern Data Pipelines	Dr. Niamh Burke	2025-05-12	300	3	297
Main Hall	Customer 360 with SQL	Dr. Niamh Burke	2025-05-14	300	3	297
Graph Lab	Scaling Neo4j for Recommendations	Prof. Alan Shaw	2025-05-12	120	3	117
Cloud Suite	Cloud Cost Optimisation	Ruth Collins	2025-05-14	180	3	177
Executive Lounge	FinTech Risk Signals	Marta Silva	2025-05-13	40	3	37
Cloud Suite	Secure API Design	Ruth Collins	2025-05-12	180	2	178
Cloud Suite	Energy Dashboards at Scale	Aisling Kerr	2025-05-15	180	2	178
Innovation Room	Building Reliable ETL Jobs	Conor Daly	2025-05-13	90	2	88
Innovation Room	EdTech Product Analytics	Grace Lennon	2025-05-15	90	2	88
Workshop Studio	Logistics Forecasting	Patrick Moore	2025-05-15	60	2	58
Graph Lab	Graph Modelling Workshop	Prof. Alan Shaw	2025-05-14	120	1	119